

One way to visualize and count the Ming-Catalan number

The name that can be named is not the enduring and unchanging name.

Tao Te Ching, Laozi

The same thing could have evolving names:

- Catalan number

- Ming-Catalan number

- Minggatu-Catalan number

The same thing can have various meanings,
and some could help with counting.

One way to visualize and count the Ming-Catalan number

n=0: 1 word

a , ,

n=1: 1 word

(ab) 'UR'

n=2: 2 words

(a(bc)) 'URUR'

((ab)c) 'UURR'

n=3: 5 words

(a(b(cd))) 'URURUR'

((ab)(cd)) 'UURRUR'

((a(bc))d) 'UURURR'

((ab)c)d) 'UUURRR'

(a((bc)d)) 'URUURR'

n=4: 14 words

(a(b(c(de)))) 'URURURUR'

((ab)(c(de))) 'UURRURUR'

((a(bc))(de)) 'UURURRRUR'

((ab)c)(de)) 'UUURRRUR'

((ab)(cd))e) 'UUURRURR'

((a(bc))d)e) 'UUURURRR'

((ab)c)d)e) 'UUUURRRR'

(a(b(cd)))e) 'UURURURR'

(a((bc)d))e) 'UURUURRR'

((ab)((cd)e)) 'UURRUURR'

(a((bc)(de))) 'URUURRUR'

(a((b(cd))e)) 'URUURURR'

(a(((bc)d)e)) 'URUUURRR'

(a(b((cd)e))) 'URURUURR'

Some more sharing. YK (2025).

One way to visualize and count the Ming-Catalan number

URURURUR

U: (_)

R: (a _)

U: (a (_))

R: (a (b _))

U: (a (b (_)))

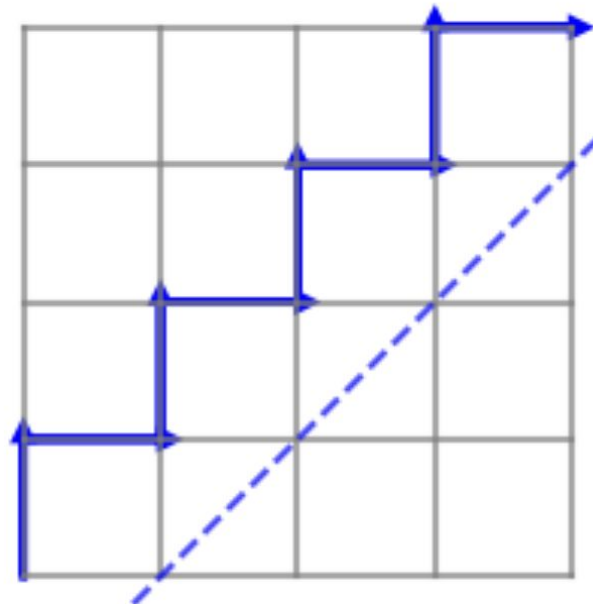
R: (a (b (c _)))

U: (a (b (c (_))))

R: (a (b (c (d _))))

Completed: (a (b (c (d e))))

(a(b(c(de))))
URURURUR



One way to visualize and count the Ming-Catalan number

UURRURUR

U: (_)

U: ((_))

R: ((a _))

R: ((ab) _)

U: ((ab) (_))

R: ((ab) (c _))

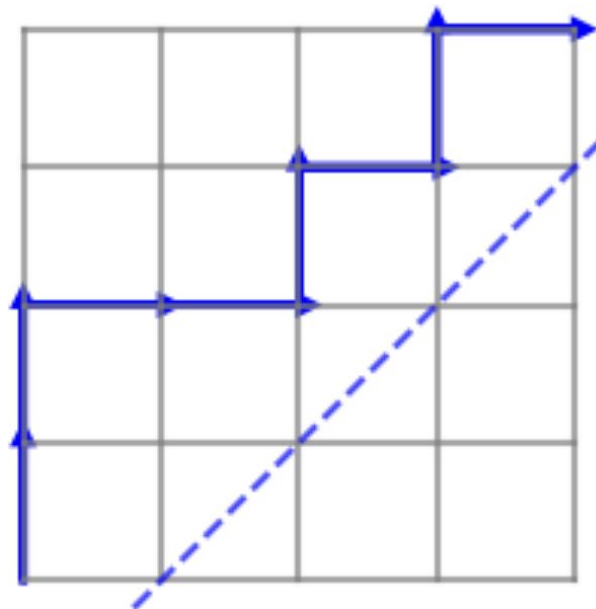
U: ((ab) (c (_)))

R: ((ab) (c (d _)))

Completed: ((ab) (c (de)))

((ab)(c(de)))

UURRURUR



One way to visualize and count the Ming-Catalan number

UURURRUR

U: (_)

U: ((_))

R: ((a _))

U: ((a (_)))

R: ((a (b _)))

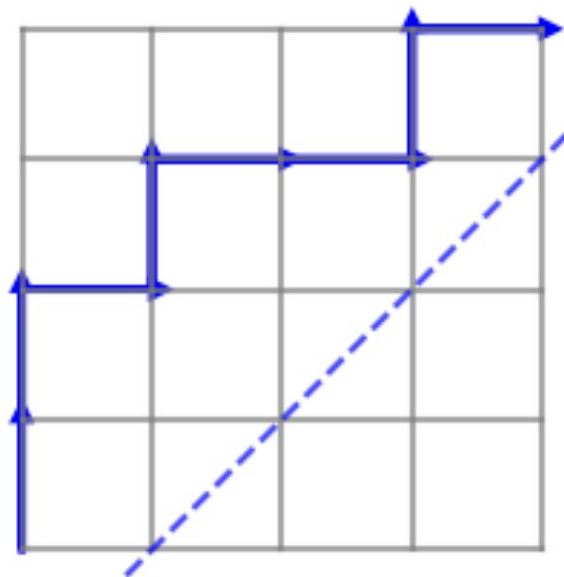
R: ((a (bc)) _)

U: ((a (bc)) (_))

R: ((a (bc)) (d _))

Completed: ((a (bc)) (de))

((a(bc))(de))
UURURRUR



One way to visualize and count the Ming-Catalan number

UUURRRUR

U: (_)

U: ((_))

U: (((_)))

R: (((a _)))

R: (((ab) _))

R: (((ab) c) _)

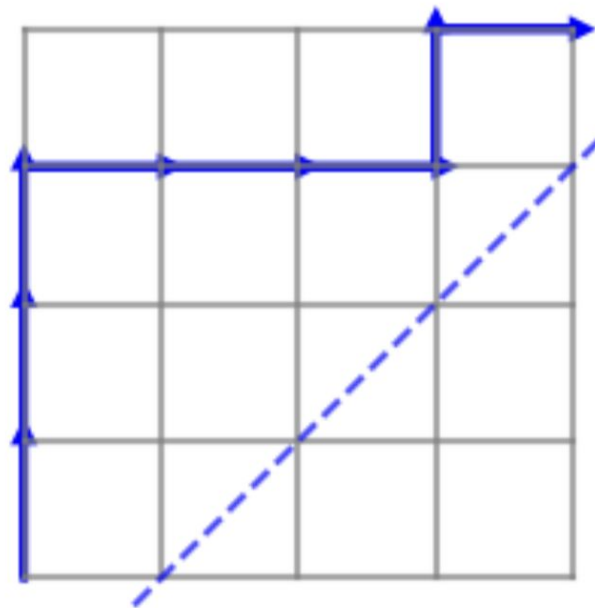
U: (((ab) c) (_))

R: (((ab) c) (d _))

Completed: (((ab) c) (de))

(((ab)c)(de))

UUURRRUR



One way to visualize and count the Ming-Catalan number

UUURRURR

U: (_)

U: ((_))

U: (((_)))

R: (((a _)))

R: (((ab) _))

U: (((ab) (_)))

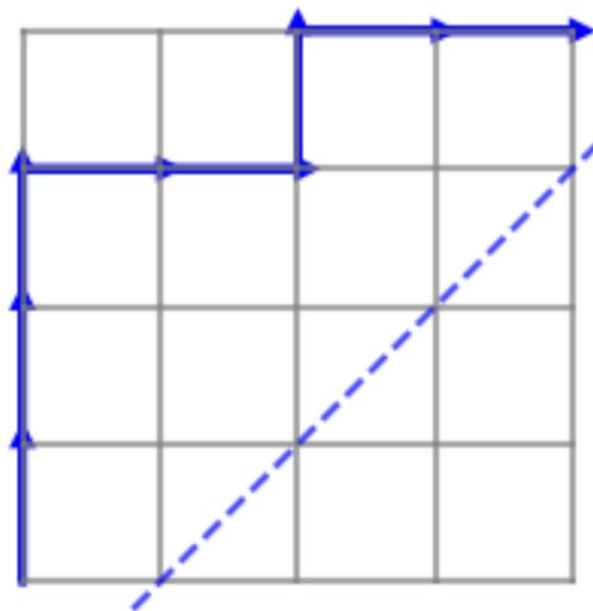
R: (((ab) (c _)))

R: (((ab) (cd)) _)

Completed: (((ab) (cd)) e)

(((ab)(cd))e)

UUURRURR



One way to visualize and count the Ming-Catalan number

UUURURRR

U: (_)

U: ((_))

U: (((_)))

R: (((a _)))

U: (((a (_))))

R: (((a (b _))))

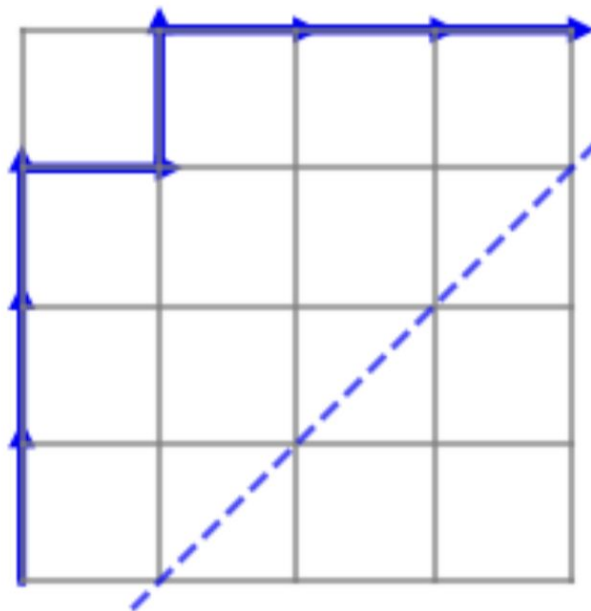
R: (((a (b c)) _))

R: (((a (b c)) d) _)

Completed: (((a (b c)) d) e)

(((a (b c)) d) e)

UUURURRR



One way to visualize and count the Ming-Catalan number

UUUURRRR

U: (_)

U: ((_))

U: (((_)))

U: ((((_))))

R: ((((a _))))

R: ((((ab) _)))

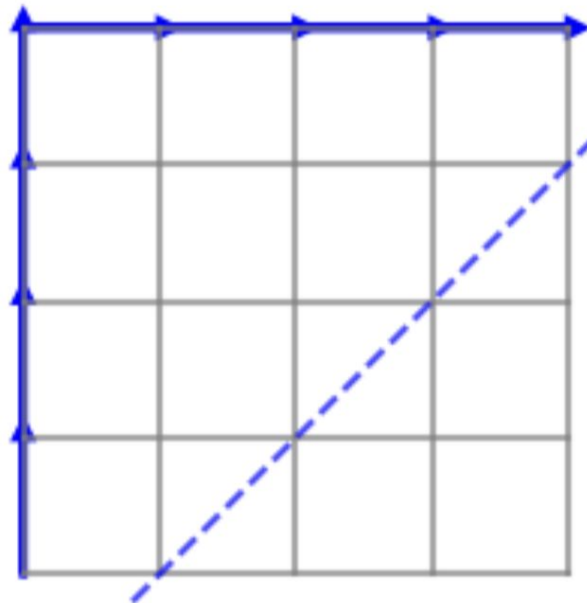
R: ((((ab) c) _))

R: ((((ab) c) d) _)

Completed: ((((ab) c) d) e)

((((ab)c)d)e)

UUUURRRR



One way to visualize and count the Ming-Catalan number

UURURURR

U: (_)

U: ((_))

R: ((a _))

U: ((a (_)))

R: ((a (b _)))

U: ((a (b (_))))

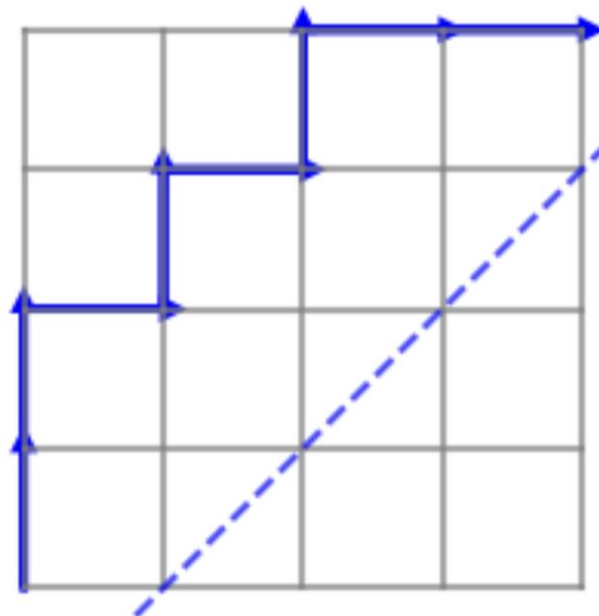
R: ((a (b (c _))))

R: ((a (b (c d))) _)

Completed: ((a (b (c d))) e)

((a(b(cd)))e)

UURURURR



One way to visualize and count the Ming-Catalan number

UURUURRR

U: (_)

U: ((_))

R: ((a _))

U: ((a (_)))

U: ((a ((_))))

R: ((a ((b _))))

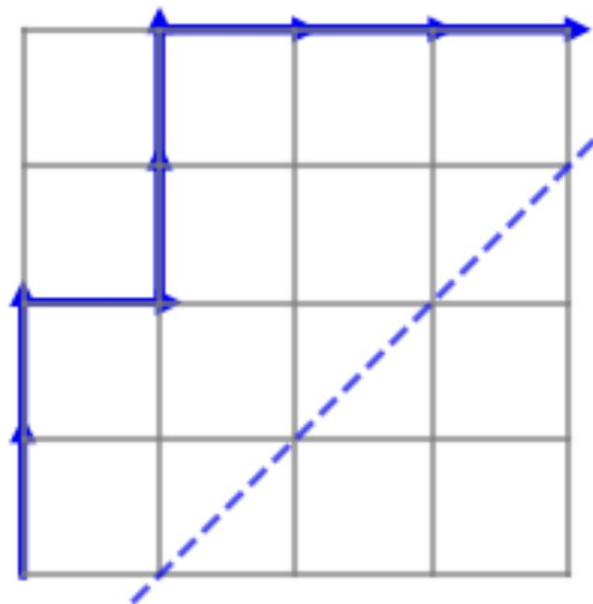
R: ((a ((bc) _)))

R: ((a ((bc) d)) _)

Completed: ((a ((bc) d)) e)

((a((bc)d))e)

UURUURRR



One way to visualize and count the Ming-Catalan number

UURRUURR

U: (_)

U: ((_))

R: ((a _))

R: ((ab) _)

U: ((ab) (_))

U: ((ab) ((_)))

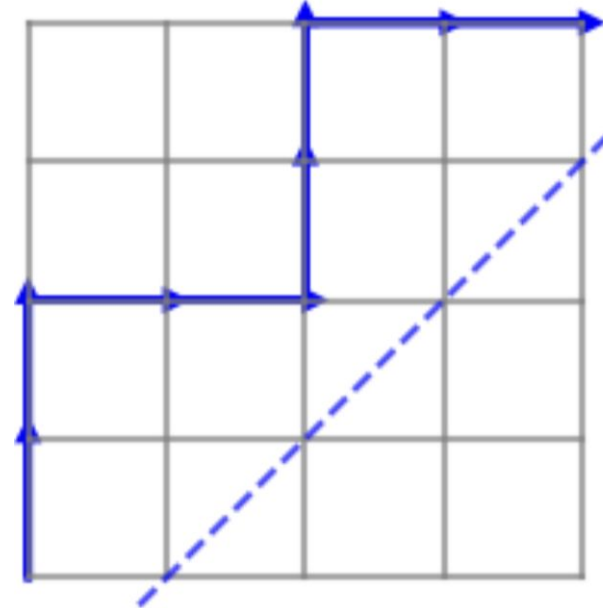
R: ((ab) ((c _)))

R: ((ab) ((cd) _))

Completed: ((ab) ((cd) e))

((ab)((cd)e))

UURRUURR



One way to visualize and count the Ming-Catalan number

URUURRUR

U: (_)

R: (a _)

U: (a (_))

U: (a ((_)))

R: (a ((b _)))

R: (a ((bc) _))

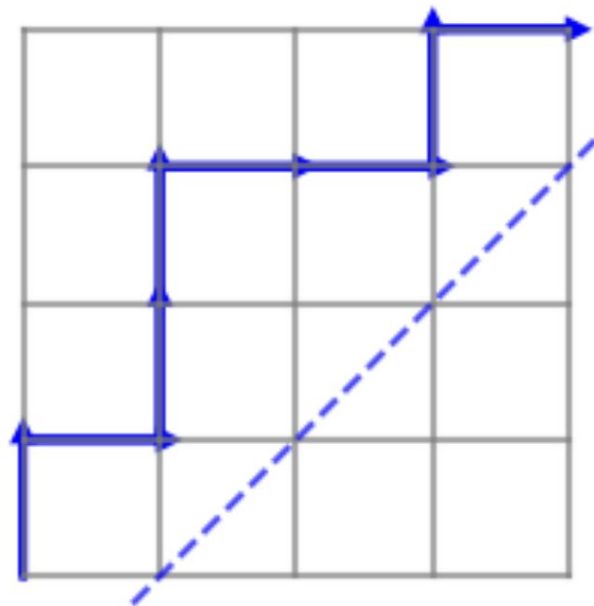
U: (a ((bc) (_)))

R: (a ((bc) (d _)))

Completed: (a ((bc) (de)))

(a ((bc) (de)))

URUURRUR



One way to visualize and count the Ming-Catalan number

URUURURR

U: (_)

R: (a _)

U: (a (_))

U: (a ((_)))

R: (a ((b _)))

U: (a ((b (_))))

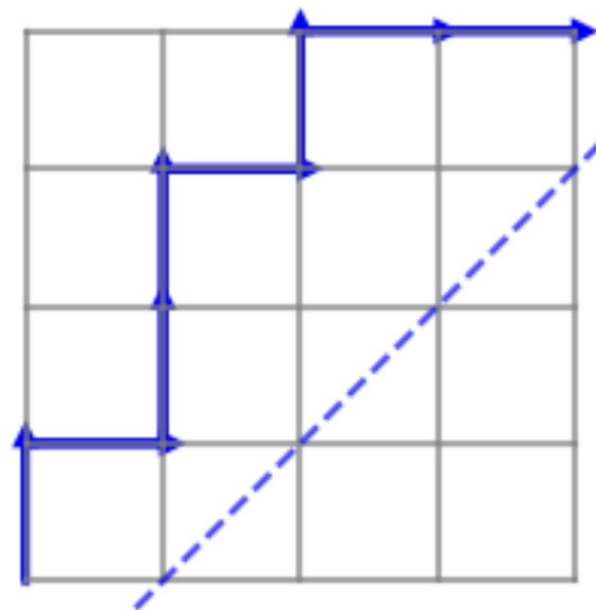
R: (a ((b (c _))))

R: (a ((b (cd)) _))

Completed: (a ((b (cd)) e))

(a ((b (cd)) e))

URUURURR



One way to visualize and count the Ming-Catalan number

URUUURRR

U: (_)

R: (a _)

U: (a (_))

U: (a ((_)))

U: (a (((_))))

R: (a (((b _))))

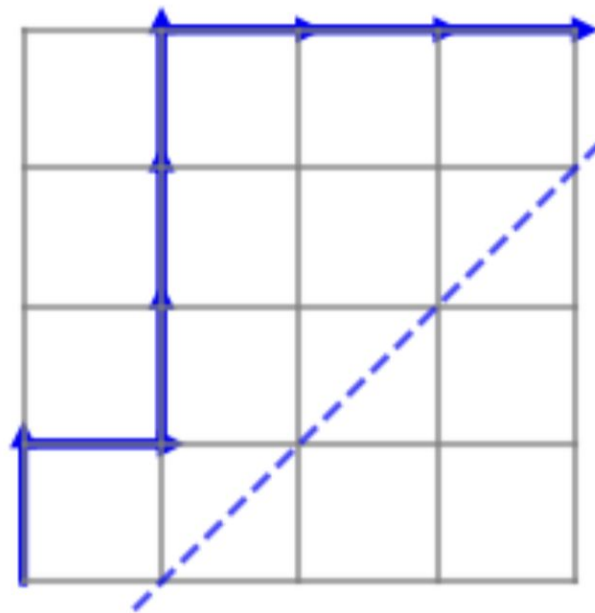
R: (a (((bc) _)))

R: (a (((bc) d) _))

Completed: (a (((bc) d) e))

(a (((bc) d) e))

URUUURRR



One way to visualize and count the Ming-Catalan number

URURUURR

U: (_)

R: (a _)

U: (a (_))

R: (a (b _))

U: (a (b (_)))

U: (a (b ((_))))

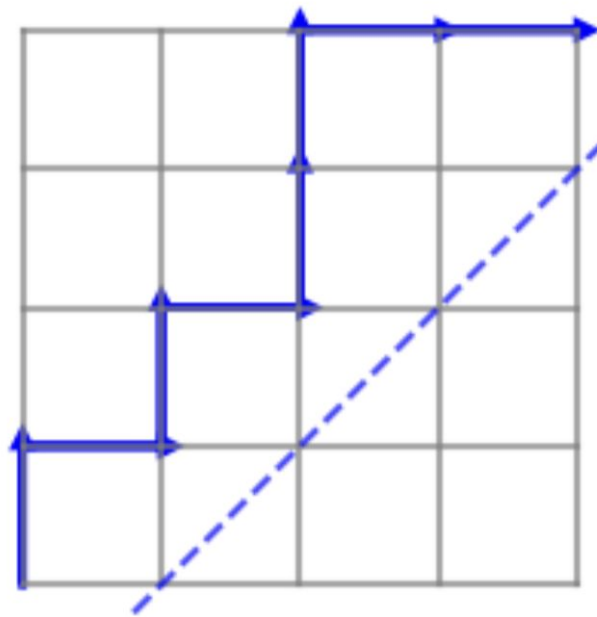
R: (a (b ((c _))))

R: (a (b ((cd) _)))

Completed: (a (b ((cd) e)))

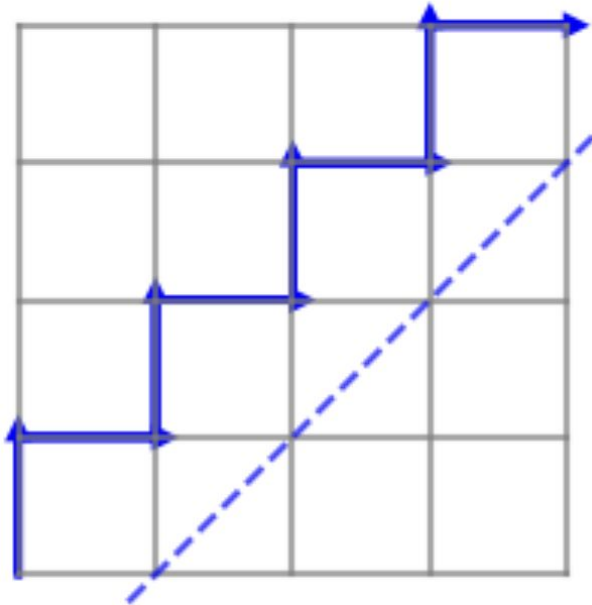
(a(b((cd)e)))

URURUURR

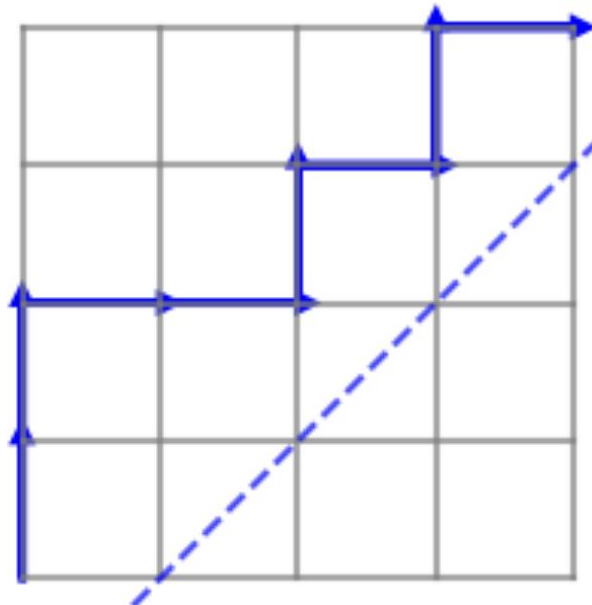


One way to visualize and count the Ming-Catalan number

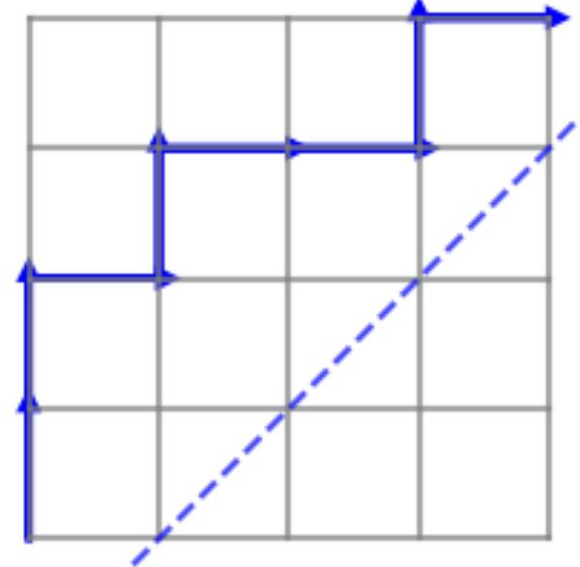
$(a(b(c(de))))$
URURURUR



$((ab)(c(de)))$
UURRURUR



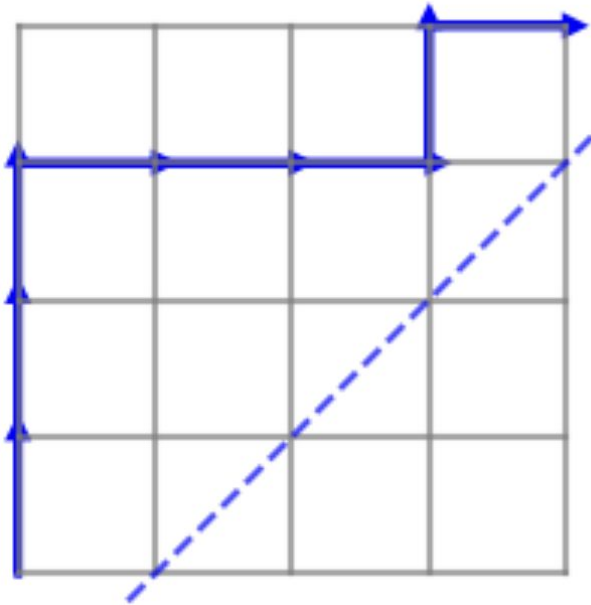
$((a(bc))(de))$
UURURRRUR



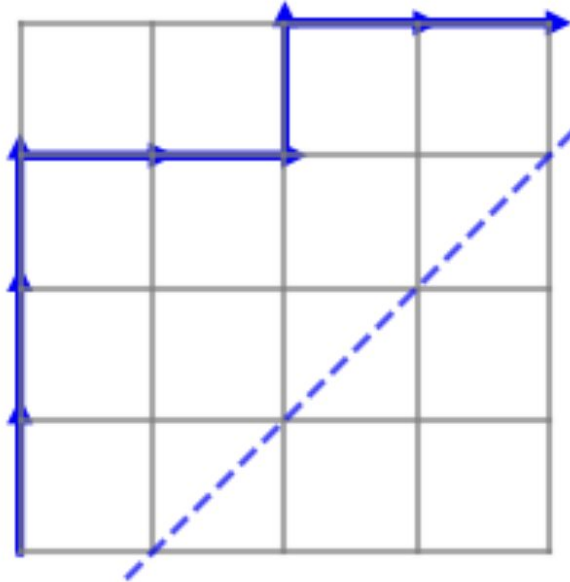
Some more sharing. YK (2025).

One way to visualize and count the Ming-Catalan number

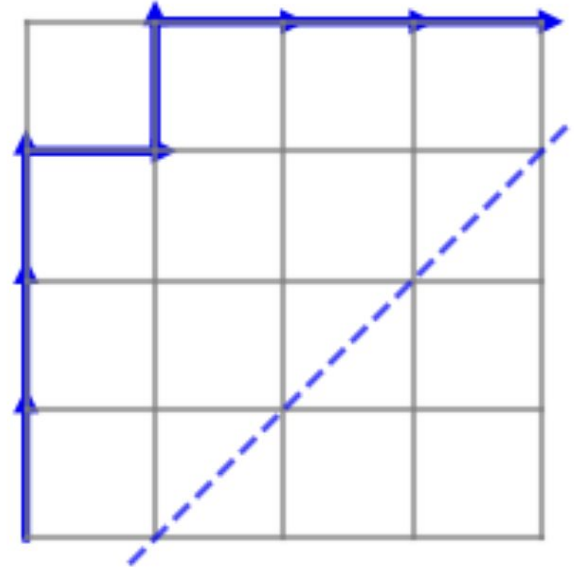
$((ab)c)(de)$
UUURRRUR



$((ab)(cd))e$
UUURRURR



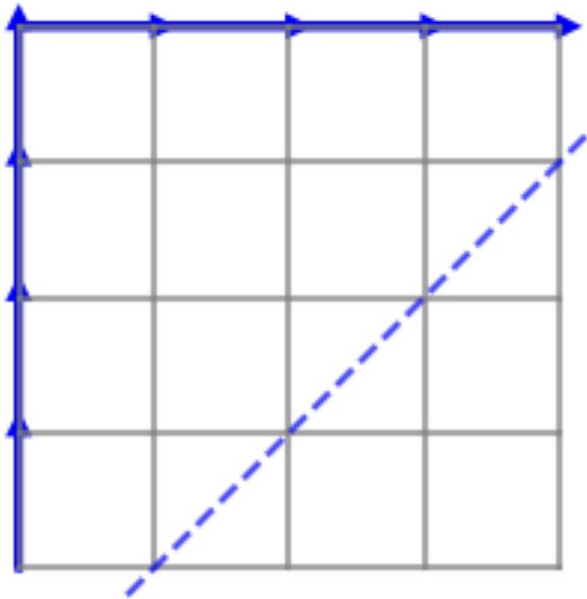
$((a(bc))d)e$
UUURURRR



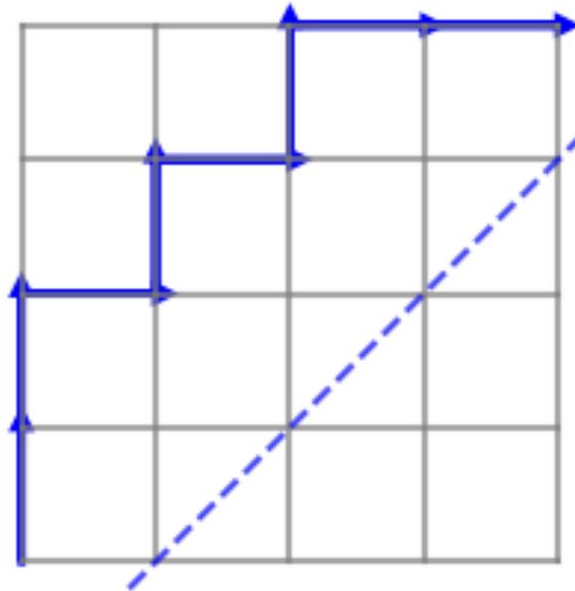
Some more sharing. YK (2025).

One way to visualize and count the Ming-Catalan number

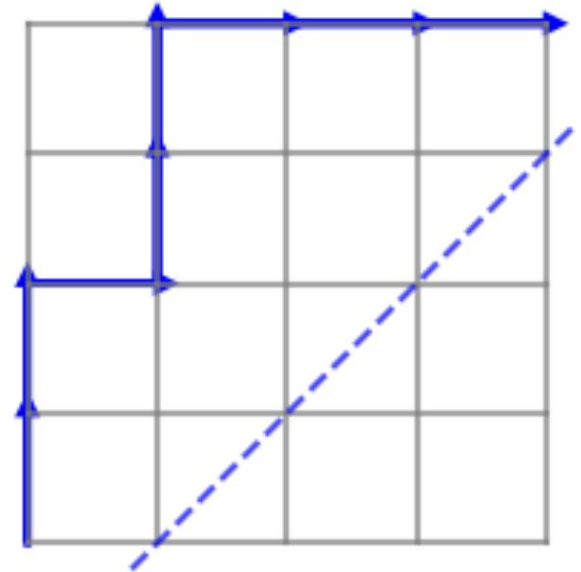
$((((ab)c)d)e)$
UUUUURRRR



$((a(b(cd)))e)$
UURURURR



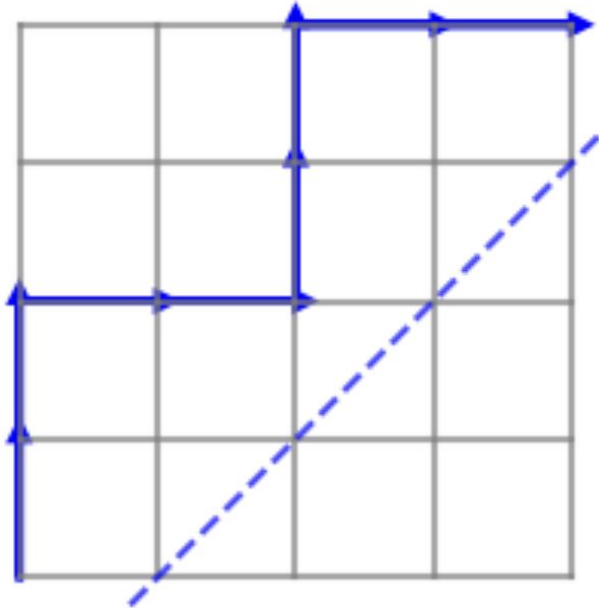
$((a((bc)d))e)$
UURUURRR



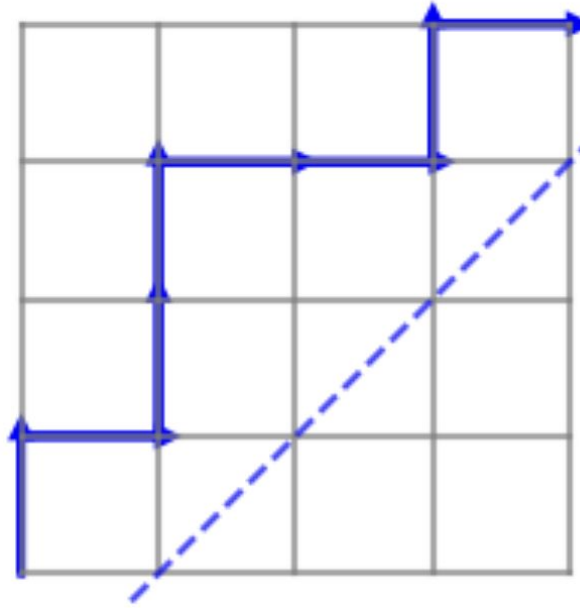
Some more sharing. YK (2025).

One way to visualize and count the Ming-Catalan number

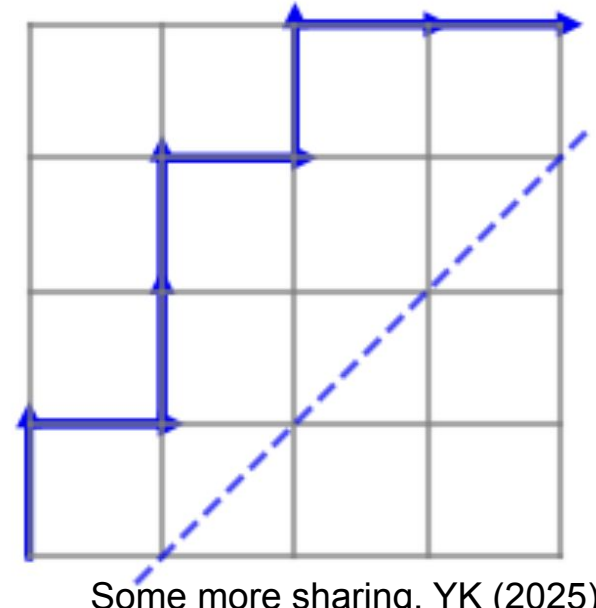
$((ab)((cd)e))$
UURRUURR



$(a((bc)(de)))$
URUURRUR



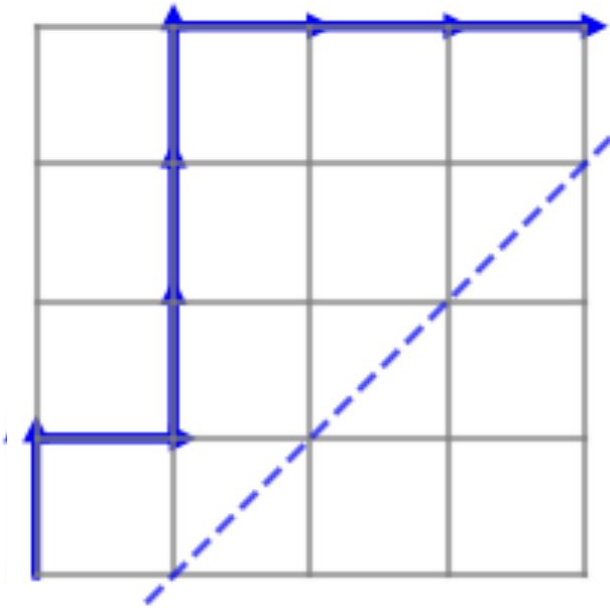
$(a((b(cd))e))$
URUURURR



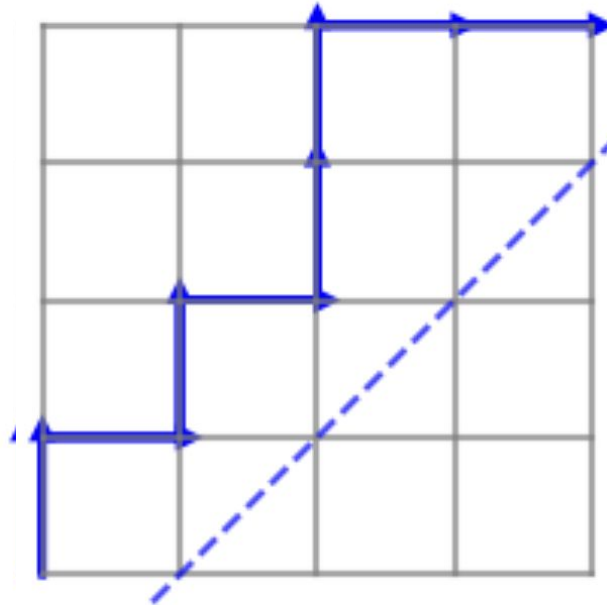
Some more sharing. YK (2025).

One way to visualize and count the Ming-Catalan number

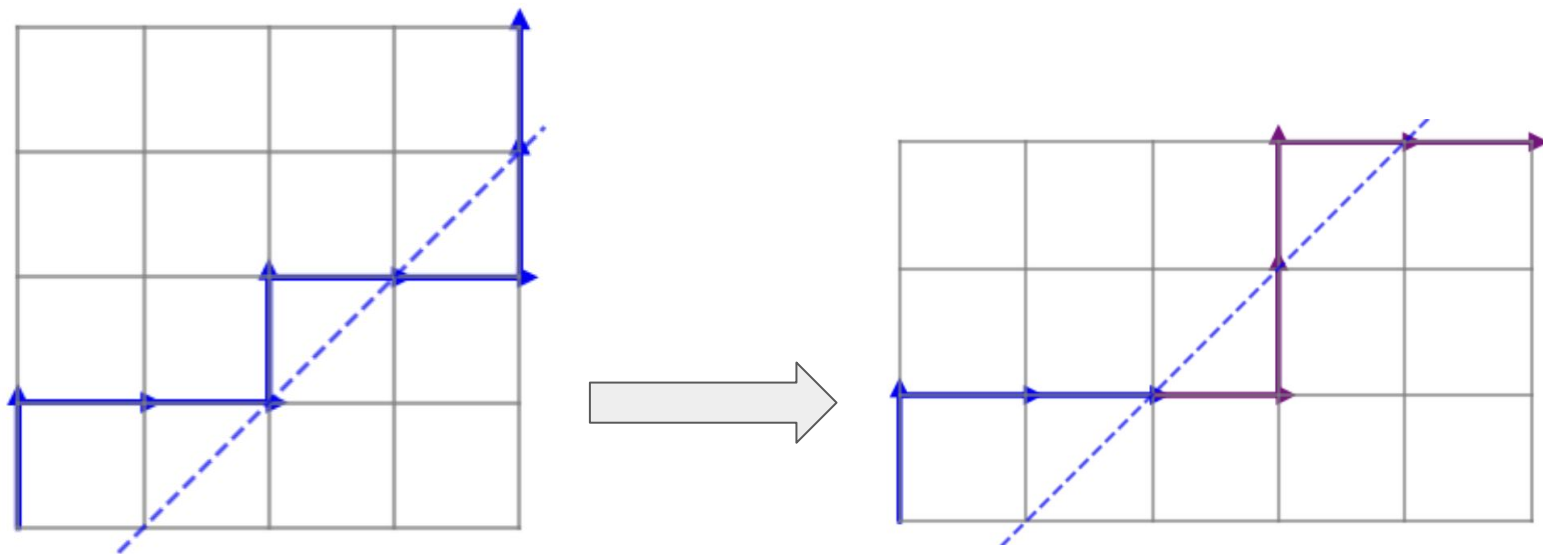
$(a(((bc)d)e))$
URUUURRR



$(a(b((cd)e)))$
URURUURR



One way to visualize and count the Ming-Catalan number



$$C_n = C_n^{2n} - C_{n-1}^{2n} = \frac{(2n)!}{n!(n-1)!} \left(\frac{1}{n} - \frac{1}{n+1} \right) = \frac{(2n)!}{(n+1)!n!} = \frac{C_n^{2n}}{n+1}.$$

Some more sharing. YK (2025).

Some history of me: some toolkits developed in 2023

第二節・語法規則與上文下理

比起分詞 (segmentation) 更精緻的思量是結合性 (association)。語法規則中，兩個字結合而成的詞彙只有單純的一種結合方式，但更多字的結合組合就會逐漸變得有趣起來。例如給定一個由 3 個字組成的分詞，語法規則中就可以有 2 種不同的結合組合；給定一個由 4 個字組成的分詞，語法規則中就可以有 5 種不同的結合組合；一般而言，一個由 $n+1$ 個字組成的分詞，語法規則中就可以有 $\frac{(2n)!}{n!(n+1)!} = \frac{C_n^{2n}}{n+1}$ 種不同的結合組合；結合組合的計算由東方的數學家明安圖和西方的數學家卡塔蘭 (Catalan) 分別在不同場景中獨立進行，所以我們可以把結合組合數目 $\frac{C_n^{2n}}{n+1}$ 稱作明安圖-卡塔蘭數。對數學有濃厚興趣的同學可以在[互聯網上找到相關的資料](#)，我們僅在以下作出少許示範：

以 3 個字「牛肉湯」開始，有 $\frac{C_2^4}{3} = \frac{6}{3} = 2$ 種結合方法，我們引入一對括號表示結合關係為 (牛肉)湯 和 牛(肉湯)；它們分別可以代表用牛肉煮的湯，以及給牛喝的肉湯。如果先驗知識 (prior knowledge) 告訴我們牛是素食動物，我們便可以再進一步推測牛肉湯是指 2 種結合方法中的第一種結合方法 (牛肉)湯。

再考慮 4 個字「牛肉湯麵」，有 $\frac{C_3^6}{4} = \frac{20}{4} = 5$ 種結合方法，如圖 1，我們引入兩對括號表示結合關係為 (牛肉)(湯麵)、(牛(肉湯))麵、牛(肉(湯麵))、((牛肉)湯)麵 和 牛((肉湯)麵)；它們分別可以代表用牛肉煮的湯麵、給牛喝完肉湯再用它做的麵、給牛吃的用肉煮的湯麵、用牛肉做湯再用這些湯煮麵、以及給牛吃的用肉湯煮的麵。再次使用先驗知識：牛是素食動物；便可以推測牛肉湯麵應該指第一種結合方法 (牛肉)(湯麵) 或第四種結合方法 ((牛肉)湯)麵；這兩種結合的分別在於前者我們會期望有牛肉可以拌麵，後者則可能沒有牛肉而只有牛肉湯和麵食了，見圖 2。在編者的認知中，一般的牛肉湯麵是有牛肉的，即是 5 種結合方法中的第一種結合方法 (牛肉)(湯麵)。

○ 實例 ● 符號

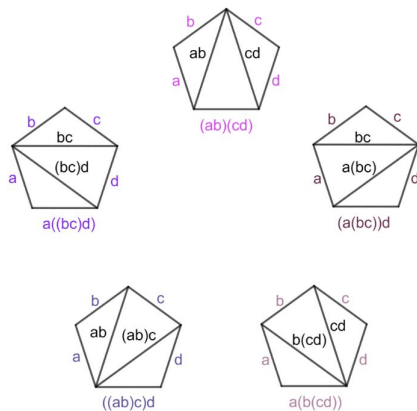


圖 1: 抽象化及可視化結合過程。如圖所示，我們可以把 $n+1$ 個字加上 $n-1$ 對括號的結合方法，對應為把凸 $n+2$ 邊形分成 n 個三角形的做法；那麼結合數目也就可以翻譯成把凸 $n+2$ 邊形分成 n 個三角形的組合數問題了。圖中列舉 $n=3$ 時的情況，即是 $n+1=4$ 個字引入兩對括號的 $\frac{C_3^{2n}}{n+1} = \frac{C_3^6}{4} = \frac{20}{4} = 5$ 種結合方法。